

Precision, Recall & F1-Score

Building Real Intuition for Classification Metrics

AWS Machine Learning Associate • Part 18

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What We'll Cover



A spam filter example



Precision



Recall



The tradeoff between them



F1 and F-beta



Multi-class & SageMaker

Our Example: A Spam Filter

	Predicted Spam	Predicted Not Spam
Actual Spam	**30** True Positives	**10** False Negatives
Actual Not Spam	**20** False Positives	**140** True Negatives

200 emails

- **40 are actually spam**
  The filter found 30 and missed 10
- **160 are legitimate**
  It passed 140 and flagged 20 by mistake
- **50 flagged as spam**
  30 of them really were spam

Spam is the positive class. Dark cells are correct predictions.

Precision

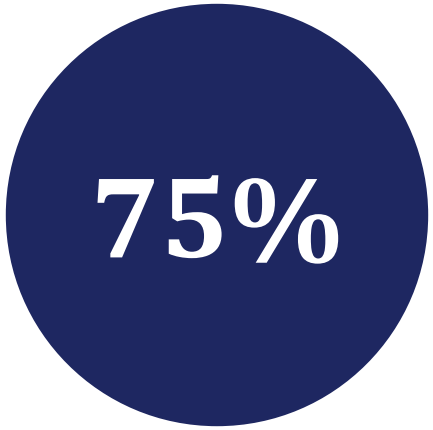


Precision

- **The question**
  When the model says "spam", how often is it right?
- **The formula**
  `TP / (TP + FP)`
- **Our example**
  30 correct out of 50 flagged = 60%

Cares about false positives (false alarms). Low precision means real emails get buried in the spam folder. High precision means you can trust the flags.

Recall



Recall

- **The question**
  Of all the real spam, how much did the model find?
- **The formula**
  `TP / (TP + FN)`
- **Our example**
  30 found out of 40 actual spam = 75%

Cares about false negatives (misses). Low recall means spam keeps slipping into the inbox. Also called sensitivity or the true positive rate.

The Precision-Recall Tradeoff

Flag Only the Surest Email

One flag, and it is right

Precision

≈ 100%

Recall

2.5%



Your Threshold

Slides you between the two extremes, so you always choose a balance

Flag Every Email

Catches all the spam

Recall

100%

Precision

20%

No free lunch: raising one usually lowers the other.

F1 Score: The Harmonic Mean

Combine precision & recall into one number

$$F1 = 2 \times P \times R / (P + R)$$

Our filter

$$2 \times 0.60 \times 0.75 / 1.35 = 66.7\%$$

Also equals 2TP / (2TP + FP + FN)

Why not a simple average?

Precision 100%, Recall 2%

Simple average **51%**

F1 score **3.9%**

F1 is high only when both are high.

F-Beta: When One Matters More



F0.5 (β = 0.5)

Precision counts more. Suits spam filtering and recommendations, where false alarms are costly



F1 (β = 1)

Precision and recall count equally



F2 (β = 2)

Recall counts more. Suits disease screening and safety checks, where misses are costly

β sets how much more you value recall than precision. Larger β favors recall.

Multi-Class: Macro, Micro & Weighted



Macro

The plain average across classes. Every class counts equally, great for seeing how you do on rare classes



Micro

Pool all predictions first, then calculate a single score



Weighted

Average across classes, weighting each by how many examples it has

Imbalanced classes? Look at macro or per-class results, not just the overall number.

Tuning for These Metrics in SageMaker



Linear Learner Model Selection

Pick the best model by accuracy, F-beta, precision at a target recall, or recall at a target precision



Set a Target

For example: at least 90% recall, then the highest precision possible



Tuning Objectives

Automatic model tuning can optimize validation precision, recall or F-beta



Imbalanced Data

Pair these metrics with resampling or class weights

Recap

- Precision: how trustworthy the positive predictions are.
- Recall: how many of the real positives you found.
- F1 balances the two; F-beta lets you favor one. Choose by the cost of mistakes.

Thanks for watching!